

Accessible AI Safeguards for Older Adults: Real-Time Scam Interception, Cognitive-Pause Verification, and Embedded Digital Literacy

Threat Landscape: Socio-Technical Manipulation and Cognitive Vulnerabilities in Elder Fraud

Financial exploitation targeting older adults represents an acute crisis within modern telecommunications, sitting at the convergence of consumer network vulnerabilities, generative deception, and age-associated cognitive changes. Statistical reporting from the Federal Bureau of Investigation (FBI) Internet Crime Complaint Center (IC3) indicates that victims aged sixty and older experienced financial losses surpassing \$3.1 billion across 88,262 recorded complaints in 2022, representing an 84% surge compared to the preceding annual period, while cumulative losses across all elder fraud categories rose beyond \$20 billion by 2025¹. The average economic harm per incident exceeds \$35,000, with thousands of individuals suffering catastrophic losses exceeding \$100,000 via irrevocable wire transfers, prepaid retail gift card draining, cryptocurrency terminal deposits, and home equity encumbrances¹.

The underlying vulnerability of older demographics does not arise from uniform intellectual degradation, but rather from sophisticated socio-technical schemes engineered to exploit specific affective, social, and physiological traits². Adversaries systematically construct pretexts that generate severe emotional arousal, thereby circumventing standard critical scrutiny².

Scam Modality	Primary Technical Vectors	Emotional and Social Triggers	Exploited Cognitive Heuristics	Financial and Psychological Severity
Government Impersonation	Spoofed Caller ID, VoIP auto-dialers, SMS lures ¹	Fear of arrest, institutional intimidation, urgent legal peril ²	Authority bias, social compliance norms, threat loss aversion ³	Severe; complete asset liquidation to "safe government accounts" ²
Grandparent / Family	AI voice cloning, SMS	Panic, acute protective	Familial affection	High; irreversible

Emergency	distress texts, Vishing ⁹	instinct, hyper-urgency ⁹	heuristic, immediate empathy, cognitive tunneling ⁹	instant payments, money transfer apps, retail gift cards ²
Tech Support Fraud	Browser pop-ups, remote access trojans (RATs), inbound phone calls ³	Perceived device compromise, loss of digital connectivity, confusion ⁴	Delegated technical authority, fear of financial data breach ³	High frequency; fraudulent service subscriptions, credential theft, wire drain ¹
Romance / Confidence Fraud	Social messaging platforms, SMS wrong-number lures ¹	Chronic loneliness, emotional dependency, curated intimacy ¹¹	Reciprocity heuristic, sunk-cost fallacy, isolation reinforcement ²	Catastrophic; prolonged multi-month grooming, complete net worth depletion ¹
Lottery / Sweepstakes / Refund	SMS notifications, automated phone calls, counterfeit checks ³	Excitement, unexpected financial relief, manufactured urgency ³	Greed/reward priming, availability heuristic, status quo bias ⁷	Moderate to high; advanced upfront fee payments, overpayment wire transfers ¹

The psychological architecture of elder fraud maps directly to dual-process decision theories, notably Daniel Kahneman's formulation of System 1 (fast, automatic, intuitive, emotionally governed) and System 2 (slow, deliberate, analytical, computationally demanding)⁷. Human cognition naturally defaults to System 1 heuristics to preserve metabolic and executive resources¹⁵. System 1 functions fundamentally as an affirmative belief engine, accepting incoming assertions at face value, whereas System 2 manages the cognitively taxing labor of doubting, interrogating, and falsifying incoming claims¹⁷.

Fraud syndicates systematically provoke intense emotional arousal—such as panic over an imminent federal arrest warrant, distress regarding an injured grandchild, or fear of compromised bank accounts—to trigger acute fight-or-flight reactions². Neurologically, elevated amygdala activation dampens prefrontal executive functioning, disabling System 2 deliberate processing¹⁴. The perpetrator enforces strict time compression and behavioral

isolation, warning victims against disconnecting the line or conferring with external confidants, which prevents targets from re-engaging analytical checks or verifying claims independently². Normal age-related cognitive changes, including moderate declines in working memory capacity and visual-auditory processing speed, compound this vulnerability under time-pressured conditions¹². Furthermore, social determinants such as social isolation and bereavement markedly heighten susceptibility, with empirical research demonstrating a

significant correlation ($r = 0.92$) between acute loneliness in older adults and vulnerability to fraudulent solicitations¹². Compounding the damage, elder victims frequently conceal victimization due to profound shame, embarrassment, or fear of losing personal independence, which inhibits timely mitigation and allows illicit networks to circulate their contact information on secondary re-victimization lists⁶.

Technical Evaluation of Current Telephony Defenses and System Deficiencies

Modern consumer communication platforms deploy defensive filtering at both the telecommunications carrier level and the mobile operating system level, yet both tiers present fundamental architectural shortcomings when shielding older demographics against conversational social engineering²¹.

Telecommunications Carrier Infrastructure: STIR/SHAKEN Limitations

The telecommunications industry, under regulatory mandates such as the United States TRACED Act, deployed the STIR/SHAKEN framework (Secure Telephone Identity Revisited / Signature-based Handling of Asserted information using toKENs) to combat illegal caller ID spoofing²⁴. Operating within IP-based Session Initiation Protocol (SIP) signaling, STIR/SHAKEN appends an encrypted digital signature (Identity Header) containing an attestation rating assigned by the originating service provider²⁴. Under this standard, Level A (Full Attestation) establishes that the carrier verifies both the caller's identity and their authorization to use the telephone number, Level B (Partial Attestation) verifies caller identity without confirming number ownership, and Level C (Gateway Attestation) denotes international or transit gateway traffic where neither identity nor number origin can be validated²².

Despite widespread implementation, STIR/SHAKEN presents fundamental conceptual and operational gaps²³. The protocol authenticates the originating carrier's relationship to the phone number; it does not evaluate, inspect, or authenticate the caller's real-time conversational intent²². Fraud syndicates systematically bypass this control by leasing or purchasing legal telephone blocks directly through complicit or loosely regulated VoIP providers, thereby securing Level A attestation for malicious voice traffic²².

Furthermore, STIR/SHAKEN does not authenticate Calling Name (CNAM) data, allowing bad actors to display legitimate organizational names such as institutional fraud divisions over properly signed lines²³. Cross-border calls and legacy non-IP (SS7) interconnections regularly degrade signatures to Gateway C attestation or strip tokens altogether, allowing spoofed calls to terminate on consumer handsets with ambiguous labeling²³.

Client-Side Defensive Applications and Operating System Features

Third-party filtering applications such as Truecaller rely on distributed, crowdsourced blacklists and historical spam reports²¹. These databases are inherently backward-looking, failing to protect against rotating VoIP numbers, targeted spear-phishing, or zero-day synthetic voice attacks²¹. Operationally, these applications impose visual complexity, display aggressive upsell paywalls, and present invasive privacy demands by harvesting address books, creating severe usability barriers for older users²¹.

On-device operating system interventions have begun integrating artificial intelligence for real-time conversation screening³¹. On-device scam detection systems on modern consumer hardware utilize compact models, such as Gemini Nano, to parse ongoing call transcripts locally and identify suspicious linguistic patterns³¹. Upon recognizing linguistic fraud markers, the system triggers high-salience visual alerts and haptic notifications, instructing the user that the call is suspected fraud and recommending that they hang up³⁴.

Defensive Paradigm	Mechanism of Operation	Primary Detection Signals	Core Strengths	Operational Failure Modes for Older Demographics
STIR/SHAKEN Protocol	Carrier SIP identity headers and public-key digital certificates ²⁴	Cryptographic attestation level (A, B, C) ²⁴	Eliminates simple caller ID spoofing on compliant IP paths ²⁵	Verifies origin, not intent; easily bypassed via purchased numbers or rogue carriers ²²
Crowdsourced Call Blockers (e.g., Truecaller)	Client-side lookup against centralized reputation databases ²¹	Historical user reports, call velocity metrics ²¹	Comprehensive block lists for high-volume automated robocalls ²¹	Ineffective against zero-day numbers; excessive advertising, high visual clutter, data privacy risks ²¹
OS Real-Time Alerting (e.g.,	On-device SLM live	Linguistic triggers	Strong privacy guarantees;	Abrupt binary warnings

Pixel / Galaxy AI)	transcription and semantic pattern matching ³¹	(demands for wire transfers, gift cards, passwords) ³²	detects novel calling numbers dynamically ³²	("Hang Up Now") induce acute panic, denial, or alert fatigue ³⁴
Assistive Cognitive-Pause System (Proposed)	On-device SLM real-time screening, contextual prompt generation, and teach-back loop	Multi-turn conversational structure, pressure tactics, dynamic risk thresholds	Preserves dignity; converts passive panic into structured, interactive verification	Requires elevated OS integration permissions and low-latency local inference

The fundamental limitation across current commercial implementations resides in their rigid, non-interactive warning architecture³⁴. Flashing red alert banners commanding a user to sever the connection immediately reproduce the exact high-stress, authoritarian framing exploited by the fraudster³⁴. When an older adult is under the influence of an active social engineering script, an abrupt warning frequently provokes cognitive denial, confusion, or dismissal, leaving them uncertain about whether the caller or the phone itself is malfunctioning⁴. Existing platforms treat defensive security as a blunt operational override rather than an accessible, cognitive mediation process.

Human-Computer Interaction and Ergonomic Architecture for Older Adults

Developing effective protective interfaces for older adults requires addressing age-related cognitive, sensory, and motor shifts while strictly respecting user agency and dignity¹⁰.

Cognitive Ergonomics: Transitioning from System 1 to System 2

Human-Computer Interaction (HCI) research demonstrates that under acute anxiety, working memory constriction impairs an older adult's ability to cross-reference contradictory information¹². Scammers deliberately weaponize this dynamic to keep the target locked in an emotionally reactive System 1 state⁷.

The primary objective of an assistive AI defense is not to seize unilateral control of the handset, but to introduce a structured "Cognitive Pause"⁷. The Cognitive Pause acts as a non-threatening psychological wedge that interrupts the caller's pacing, lowers affective arousal, and recruits deliberative System 2 processing⁷. By converting an alarming, ambiguous threat into an orderly series of simple actions, the interface restores agency to the user¹⁰.

Accessibility and Multi-Modal Interface Parameters

Digital interfaces designed for older adults must mitigate sensory degradation, including

presbyopia, reduced contrast sensitivity, and high-frequency hearing loss, alongside motor challenges such as tremors and diminished tap precision¹⁹. Interfaces must strictly adhere to WCAG AAA contrast compliance, avoiding saturated, rapid flashing red banners that provoke visual disorientation or acute stress³⁴. Graphic surfaces benefit from calm, high-contrast palettes, utilizing dark slate backdrops paired with warm amber indicators for cautionary states and pure white for primary content, accompanied by sans-serif typefaces rendered at a minimum scale of 18 to 20 physical points.

Interactive elements must provide generous physical touch targets, expanding beyond standard specifications to bounds of 48 to 64 density-independent pixels, surrounded by clear spacing to eliminate accidental inputs. Information density must remain low, presenting one discrete conversational recommendation per screen to avoid splitting user attention⁴¹. Auditory prompts should avoid high-frequency warning sirens in favor of low-frequency melodic chimes within the 500 Hz to 1.5 kHz range, while tactile alerts rely on steady, rhythmic double pulses that signal attention without mimicking urgent device error codes²¹.

Designing for Dignity and Autonomy

A primary reason older adults abandon assistive technologies is the patronizing tone embedded in protective software¹⁰. Applications that label the user as frail or vulnerable generate acute psychological resistance, leading users to reject alerts to defend their personal autonomy¹¹. Interfaces must position the AI system as an administrative concierge or an executive assistant verifying procedural facts, rather than a guardian evaluating user competence¹⁰. Defensive copy must avoid alarming declarations in favor of calm, neutral observations, framing verification steps as standard institutional operating procedures rather than emergency measures⁷.

System Engineering: End-to-End On-Device Detection and Verification Architecture

Implementing real-time conversational analysis on consumer mobile hardware requires balancing operating system permissions, inference latency, hardware resource constraints, and telecommunications privacy laws³².

Operating System Ingestion Pipeline: Android and iOS Telephony Constraints

Capturing and processing incoming SMS, MMS, and real-time voice calls requires deep operating system integration, which varies considerably across mobile platforms⁴⁵.

On the Android platform, initial pre-call evaluation can be established through the CallScreeningService API⁴⁵. Upon receiving an incoming telephony event, the platform triggers the onScreenCall(Call.Details) method, exposing metadata such as caller ID and presentation settings⁴⁵. However, CallScreeningService operates under strict system constraints, mandating that the service return a respondToCall() decision within five seconds, while limiting its authority to allowing, rejecting, or silencing the incoming ring⁴⁵. It cannot draw persistent interactive UI elements over active calls or access live conversational voice streams once the

connection is established⁴⁵.

To analyze conversational audio and present interactive, context-aware verification prompts during a call, the software must implement `InCallService` and secure the `RoleManager.ROLE_DIALER` privilege⁵⁰. Assuming the default dialer role grants complete programmatic control over the call lifecycle⁵⁰. This enables the application to render heads-up window overlays via system accessibility permissions, access local audio input/output channels for real-time speech transcription, and display contextual prompt cards directly above the dialer interface⁵⁰.

On iOS, third-party applications face strict sandbox boundaries that prevent background call audio capture⁴⁷. However, short-message communications can be processed via Apple's `IdentityLookup` framework⁴⁷. Developers subclass `ILMessageFilterExtension` and implement the `ILMessageFilterQueryHandling` protocol⁴⁷. When an SMS or MMS arrives from an unsaved number, the system routes the sender metadata and message payload to the `handle(queryRequest:context:completion:)` method⁴⁷. Because the operating system isolates the extension from network access and prevents writing to shared group storage, all classification and prompt-generation logic must execute locally using models bundled within the extension itself⁴⁷.

Edge Machine Learning Runtime and Quantization

Real-time conversational protection requires local inference to ensure low latency and user privacy³². Cloud-hosted language models introduce round-trip network delays, vulnerability to poor cellular connections, ongoing infrastructure costs, and significant data privacy risks³².

The client architecture pairs on-device Small Language Models (SLMs) with specialized edge runtimes, such as Google LiteRT-LM, which succeeded the earlier MediaPipe LLM Inference framework⁵⁵. Highly compact architectures, including Gemma 2B, Gemma 3n, and Gemma 4 E2B, operate efficiently on mobile hardware when quantized to 4-bit precision via dynamic range quantization, leveraging XNNPack runtimes and ARM v9 I8MM vector instructions⁵⁵. This optimization compresses the base model into an active RAM footprint of 1.5 to 2.2 gigabytes while preserving the semantic reasoning needed to identify social engineering tactics⁴⁶.

Data processing follows a low-latency, two-stage inference cascade. Incoming conversational audio is converted to text using an on-device Automatic Speech Recognition (ASR) engine, such as a quantized Whisper-tiny model or the native platform speech recognizer, feeding a sliding circular context window³². In the first stage, a lightweight classifier combining regular expression heuristics with an int8 MobileBERT model continuously monitors the transcript for manipulative patterns, such as demands for one-time passwords, prepaid card payments, remote desktop software, or urgent legal threats³. In the second stage, if the intent score crosses a preset risk threshold, the accumulated conversation history is forwarded to the on-device Gemma SLM⁵⁵. The model returns a structured JSON payload containing a risk classification, a plain-language explanation of the threat, and two targeted verification questions for the user to read to the caller⁵⁵.

On current mobile chipsets equipped with Neural Processing Units (NPUs), Time to First Token (TTFT) settles between 300 and 600 milliseconds, streaming tokens at 15 to 25 tokens per

second directly into the application's reactive StateFlow architecture³².

System Layer	Android Implementation	iOS Implementation	Memory & Latency Budget	Privacy & Security Isolation
Ingestion Hook	InCallService (ROLE_DIALER) & Telephony.SMS_RECEIVED [cite: 50, 51]	IdentityLookup (ILMessageFilterExtension) ⁴⁷	< capture latency	Local OS process; sandboxed background service ⁴⁷
Speech-to-Text (ASR)	On-device Whisper-tiny or OS SpeechRecognizer API ³²	Native SFSpeechRecognizer (Offline model)	< chunk processing	In-memory circular buffer; no disk caching ³²
Semantic Extraction	MobileBERT int8 intent classifier	CoreML Quantized DistilBERT	25 MB RAM; < inference	Zero cloud data egress; fully local model weights ³²
Generative SLM Engine	LiteRT-LM running Gemma-2B / Gemma-3n int4 ⁵⁵	CoreML / LiteRT running Gemma-2B int4 ⁴⁶	1.5 — RAM; 15 — [cite: 46, 59]	Ephemeral token context window; auto-flushed post-call ⁴⁶
Assistive UI Surface	Jetpack Compose WindowManager Overlay ⁵⁰	Native Messages filter folders + Push View ⁴⁷	< frame render (60 FPS)	High-contrast visual cards; accessibility focus ⁴³

Legal, Regulatory, and Wiretapping Compliance

Analyzing live voice streams introduces important legal considerations regarding electronic surveillance and communications privacy⁶². Under United States federal law, Title III of the Omnibus Crime Control and Safe Streets Act (18 U.S.C. § 2511, the Federal Wiretap Act) establishes a one-party consent standard, permitting audio interception when one participant agrees to the process⁶². However, twelve states enforce two-party or all-party consent rules,

requiring all individuals on a call to consent before acoustic communications may be recorded or monitored⁶².

To maintain compliance with wiretap statutes and privacy frameworks such as GDPR and CCPA, the architecture operates entirely on ephemeral, in-memory processing³². Audio data is held exclusively in volatile RAM circular buffers, converted directly into text tokens, and immediately overwritten³². No audio waveforms or full call transcripts are written to persistent storage or shared with external servers³². Because local acoustic processing occurs purely within temporary memory to provide an accessibility service—without human listening or permanent records—it mirrors assistive tools like native live captioning³². For jurisdictions that interpret automated transcription as an interception event, the system includes an optional introductory chime or brief synthesized announcement upon connection, notifying the caller that assistive call verification is enabled.

Context-Aware Verification Framework: Scripted Reverse-Challenges and Out-of-Band Protocols

The primary operational innovation of the platform is shifting the older adult from a passive target into an active, empowered interrogator¹⁰. When an inbound interaction is flagged as high-risk, the model generates context-aware verification prompts designed to exploit the structural information asymmetries inherent in fraud operations².

Fraud syndicates operate from standardized, volume-driven scripts; they rarely possess detailed, private biographical knowledge about a target's family history, physical banking locations, or household specifics². Presenting targeted, difficult-to-guess questions disrupts the caller's rehearsed pacing and quickly exposes the deception, often prompting the caller to end the interaction themselves².

Scam Vector	Intercepted Threat Marker	Generated Real-Time Question for the User to Ask	Strategic Verification Mechanism	Recommended Out-of-Band (OOB) Action
Grandparent / AI Voice Clone	Caller mimics a grandchild's voice, claiming sudden arrest or an accident requiring bail ⁹ .	<i>"Before we proceed, what is our secret family safe word, or what was the name of our childhood dog on Maple</i>	Exploits the AI voice clone's lack of private biographical context; forces authentication ⁶¹ .	Disconnect immediately. Dial the grandchild's saved contact number directly ⁶¹ .

		<i>Street?"</i> [cite: 61, 65]		
Government Impersonation (IRS / DEA / SSA)	Claims Social Security number is suspended due to crime; threatens arrest ² .	<i>"What is your specific employee badge number and desk extension? I am writing both down to verify."</i> [cite: 2]	Disrupts the scammer's pacing; federal agents do not resolve criminal cases via retail gift cards or wire transfers ² .	Disconnect immediately. Contact the local police department or official agency directory number ² .
Financial / Fraud Department	Fraud alert alleging suspicious purchases; demands money transfer to "secure" funds ³ .	<i>"Which specific bank branch do you work at, and what are the last two digits of the account you are referencing?"</i>	Fraudsters rarely know verified account numbers; they rely on the mark to volunteer details ³ .	Disconnect immediately. Call the customer service number printed directly on the physical bank card ⁸ .
Tech Support Fraud (Microsoft / Apple)	Claims device is compromised; requests immediate remote access software download ³ .	<i>"What is the exact Service Ticket ID associated with my operating system serial number in your support logs?"</i> [cite: 4]	Operating system providers do not place unsolicited calls regarding local consumer device errors ³ .	Disconnect the computer from the network. Power down the device. Call a verified family member ⁴ .

These interactive scripts support the standard industry defensive guideline: **Hang Up and Verify Out-of-Band**⁸. Out-of-Band (OOB) verification terminates the untrusted inbound channel and establishes a separate, trusted connection over a known path⁸. To help older users take action, the interface pairs generated questions with clear, prominent buttons:

- A direct contact button that dials a saved family member immediately.
- A verified bank lookup button that initiates an out-of-band call to the number on their card.
- A prominent disconnect button that safely ends the call and transitions the display to a calming status screen.

Establishing a private "Family Safe Word" provides an effective defense against generative voice impersonations¹⁰. Agreed upon during a prior family check-in, the phrase is stored exclusively in encrypted on-device preferences and is never transmitted over external networks⁶¹. If a caller claiming to be a distressed relative fails to provide this passphrase, the interface advises the senior to hang up immediately and place an outbound call to the relative's verified contact entry⁶¹.

Pedagogical Mechanics: Transforming In-the-Moment Alerts into Long-Term Literacy

Traditional security education for older adults typically relies on static informational brochures, community presentations, or annual reviews, which rarely produce lasting behavioral change⁶⁶. Under the Ebbinghaus forgetting curve, individuals forget more than 50% of conceptual material within an hour, and up to 70% within 24 hours, without immediate, contextual reinforcement⁶⁷. Real-time scam interception provides a valuable, highly receptive learning opportunity⁶⁶.

Inoculation Theory (Prebunking)

Inoculation theory indicates that psychological resilience against deception can be developed similarly to biological immunity: by exposing individuals to weakened, manageable examples of manipulative techniques alongside explanations of their mechanics⁷⁰. Rather than relying entirely on reactive warnings, the system introduces low-stakes practice exercises during quiet periods⁷⁰. The application delivers occasional, opt-in simulations—such as an artificial package delivery SMS—that guide the user through identifying red flags and selecting appropriate responses, building confidence before an actual incident occurs⁷⁰.

Just-in-Time Microlearning Architecture

When a suspicious call or message is intercepted, the user experiences significant emotional relief once the encounter concludes². This post-incident window offers an optimal opportunity for contextual learning⁶⁶. Within two to five minutes of resolution—allowing sufficient time for acute stress to subside—the platform delivers a concise, 60-to-90-second learning summary⁶⁶. The learning card reviews the specific social engineering tactic encountered during the call, emphasizing three core ideas:

- The anatomy of the deception, explaining why the scammer claimed an urgent emergency or impersonated a financial institution⁸.
- The specific conversational warning sign, highlighting how pressure tactics and demands for secrecy are hallmarks of fraud².
- Positive reinforcement of the user's action, affirming that pausing and asking for verification kept them secure⁷.

The Teach-Back Method for Cognitive Reinforcement

To reinforce retention, the interface adapts the clinical "Teach-Back" method, which has demonstrated success in improving health literacy among older adults⁷⁵. Rather than asking the user to dismiss a generic acknowledgment screen, the system prompts them to answer a brief,

interactive scenario in their own words⁷⁵. Presenting a straightforward question—such as identifying the correct step to take when a caller demands gift card payments—requires the user to actively retrieve the concept, which strengthens long-term memory encoding far more effectively than passive reading⁷⁵.

Intergenerational Digital Stewardship: The "Warm Steward" Model

Complete isolation from social support networks remains a key factor in elder fraud vulnerability⁴. However, monitoring setups that expose every phone call, text message, and location pin to family members often feel intrusive and diminish personal dignity¹⁰.

The platform implements an intergenerational "Warm Steward" framework¹⁰. Instead of sharing raw transcripts or message logs, the software transmits high-level, privacy-preserving event summaries to a designated family member or caregiver only when a verified threat pattern is intercepted¹⁰.

The notification simply informs the caregiver that an impersonation attempt was intercepted and safely handled, suggesting a friendly check-in call. This approach provides family awareness without exposing private communications, transforming surveillance into supportive, respectful care⁴.

Hackathon Implementation Blueprint, System Workflow, and Pitch Strategy

End-to-End User Journey Narrative

The following practical scenario illustrates how the system operates during an encounter:

An older adult receives an incoming voice call from a local area code displaying Level C gateway attestation²⁴. The caller claims to be a federal investigator, stating that the recipient's Social Security number is linked to financial crimes and threatening immediate arrest unless their funds are transferred to a temporary government escrow account².

As the call progresses, the on-device speech-to-text pipeline transcribes the incoming audio within volatile memory³². When the caller insists that the user transfer their savings right away to avoid arrest, the intent classification model flags key warning markers: government impersonation, arrest threats, and demands for immediate fund transfers².

A subtle, low-frequency double chime plays, and a clean, high-contrast overlay card slides into view on the call screen²¹. The card clearly states that official government agencies do not suspend Social Security numbers or request phone transfers². Below the notice, the display presents two clear verification questions⁶¹:

- *"What is your official employee badge number and desk extension?"*
- *"Which local field office are you calling from?"*

The user reads the first question aloud. Unprepared for the challenge, the caller tries to regain control with renewed threats, but the user simply taps the prominent button labeled [Hang Up & Verify Official Number], ending the call².

A few minutes later, a gentle notification appears on the handset⁶⁶. Opening it displays a brief summary explaining how scammers use fake government badges to induce panic, noting that

real agencies communicate through postal mail and never demand instant phone transfers². The user taps the correct response on a quick review card, reinforcing the concept⁷⁵. Concurrently, their designated family contact receives a brief note confirming that a government impersonation call was safely handled, prompting an encouraging check-in¹⁰.

Minimum Viable Product (MVP) Technical Architecture

Developing a practical prototype within a 24- to 48-hour hackathon environment requires focusing engineering efforts on core functionality rather than building complex backend infrastructure:

The application is structured as a native Android project using Kotlin and Jetpack Compose, targeting API Level 34⁴⁶. Telephony ingestion is handled by an incoming SMS receiver paired with a mockable InCallService harness that plays pre-recorded elder fraud audio tracks and feeds them into the analysis engine².

Machine learning inference runs locally via the LiteRT-LM runtime using an int4-quantized Gemma 2B or Gemma 3n model⁵⁵. For development teams with resource-constrained test devices, a local low-latency proxy serving the same open-weights Gemma model over a local connection can simulate sub-second responses effectively⁴⁶.

The user interface uses Jetpack Compose to display the heads-up call overlay, high-contrast prompt cards, prominent disconnect controls, and the follow-up review screen⁴³.

Intergenerational updates are demonstrated via local notifications showing how family check-ins are formatted¹⁰.

Strategic Positioning and Pitch Matrix

A compelling presentation must clearly contrast the solution against existing commercial alternatives across four key dimensions:

Strategic Dimension	Legacy Telephony Blockers (e.g., Truecaller)	Native OS Call Screening (e.g., Google Pixel / Samsung)	Proposed Assistive AI Solution
Detection Timing	Pre-call only; queries static reputation lists ²¹	In-call; flags linguistic patterns via on-device SLM ³¹	In-call and in-message; continuous local semantic monitoring ³²
User Agency & Interaction	Passive; blocks or labels calls silently ²¹	Authoritarian; surfaces blunt "Hang Up" warnings ³⁴	Active; provides structured reverse-challenge questions ⁴²

Cognitive Approach	Ignores emotional factors; relies on user tech literacy ²¹	Induces stress through high-salience warning banners ³⁴	Introduces a calm "Cognitive Pause" to shift from System 1 to System 2 ⁷
Long-Term Educational Impact	None; pure reactive filtering ²¹	None; dismissible operational alerts ³⁴	Continuous; post-incident microlearning and teach-back review ⁶⁶
Family / Caregiver Model	None, or invasive full contact uploads ²¹	None; confined to the local handset	Privacy-preserving "Warm Steward" alerts for family support ¹⁰

The pitch strategy focuses on three core principles:

- Preserving dignity over rigid overrides: Rather than treating older adults as vulnerable dependants, the system provides clear, actionable prompts that turn high-pressure calls into manageable interactions while keeping the user in control¹⁰.
- A continuous educational cycle: Every intercepted call becomes a 60-second learning opportunity, using proven teach-back methods to turn passive protection into growing confidence and digital resilience⁶⁶.
- Privacy-by-design local execution: Running small models on-device within temporary memory eliminates data leakage, ensures regulatory compliance, and provides fast, cost-effective performance without cloud fees³².

Synthesis

Protecting older adults from social engineering requires moving past simple number blacklists and restrictive, paternalistic controls²³. While network protocols like STIR/SHAKEN and spam databases provide initial filtering, they cannot detect live conversational manipulation²¹. By pairing efficient on-device language models with clear, accessible interface design, the platform introduces a timely cognitive pause that helps older adults navigate stressful calls safely⁷. Combining real-time verification prompts with follow-up microlearning transforms potential threats into positive learning experiences, enabling older adults to communicate with confidence, independence, and dignity⁶¹.

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